Targeted Subsampling for Crypto Regime Models

What worked, what failed, and why online weighting is still the money idea

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The high-level question

Can we do scalable Bayesian regime inference for crypto returns?

Why this is interesting

Crypto markets switch between calm, stressed, and chaos regimes. If we can infer those regimes fast and with uncertainty, we get better risk control, drawdown avoidance, and possibly, yes, more mooolah.

Why this is hard

HMM likelihoods want full forward-backward passes. SGMCMC avoids full passes by using subsequences, but uniform subsequences miss rare/high-impact regimes.

The modeling target

- Observations: daily crypto returns.
- Latent state z_t: market regime.
- Emission model:

$$y_t \mid z_t = k \sim f_k(\theta_k).$$

- Transition:

$$z_t \mid z_{t-1} \sim \text{Categorical}(A_{z_{t-1},:}).$$

Interpretation

State 0 might be "normal market."
State 1 might be "stress/high volatility."
A good posterior answers: are we entering the bad regime, and how sure are we?

Why stochastic-gradient MCMC?

Split the time series into blocks $C_1, \ldots, C_B$:

$$\log p(\theta \mid y_{1:T}) = \log p(\theta) + \sum_{b=1}^{B} \ell_b(\theta).$$

Full gradient:

$$G(\theta) = \nabla \log p(\theta) + \sum_{b=1}^{B} g_b(\theta), \qquad g_b(\theta) = \nabla \ell_b(\theta).$$

Stochastic-gradient MCMC replaces the sum with sampled blocks:

$$\widehat{G}(\theta) = \nabla \log p(\theta) + \frac{1}{M} \sum_{m=1}^{M} \frac{g_{I_m}(\theta)}{p_{I_m}}.$$

Core design choice

Which blocks should we sample?

Uniform sampling is easy, but kind of dumb

Uniform block sampling:

$$p_b = \frac{1}{B}.$$

This is unbiased and simple.

But if most blocks are boring and a few blocks contain crashes, jumps, or rare regimes, uniform sampling spends most computation on low-value blocks.

Crypto analogy

Uniform sampling treats a sideways Tuesday and an exchange-liquidation cascade as equally informative.
That is emotionally democratic.
It is not computationally optimal.

TASS: targeted subsampling from Ou, Young, Sen, Dunson 2024

Rationale

For HMMs with rare latent states, uniform subsequences often exclude rare-state observations. Then gradients for rare-state parameters are noisy or prior-dominated.

If latent states were known, define the component-wise block gradient

$$\gamma_{n,i}(\theta) = \sum_{t \in C_n} \partial_{\theta_i} \left[\log p_\theta(y_t \mid x_t) + \log p_\theta(x_t \mid x_{t-1}) \right].$$

For component i, an unbiased one-block estimator is

$$\widehat{G}_i = \frac{\gamma_{J,i}}{a_J^{(i)}}, \qquad J \sim a^{(i)}.$$

TASS math in one slide

The second moment is

$$\mathbb{E}\{\widehat{G}_i^2\} = \sum_{n=1}^{B} \frac{\gamma_{n,i}(\theta)^2}{a_n^{(i)}}.$$

Minimize over probabilities $a_n^{(i)}$:

$$a_n^{(i),\star} \propto |\gamma_{n,i}(\theta)|.$$

Translation

Sample blocks in proportion to how much they matter for that parameter.

Practical TASS approximation

Replace unknown states by initial cluster labels, fix weights near a pilot/MAP value, and over-sample blocks with rare/extreme clusters.

Where our online model enters

Static TASS computes weights once:

$$p_b^S = (1-\lambda)\frac{\widehat{a}_b^S}{\sum_r \widehat{a}_r^S} + \frac{\lambda}{B}.$$

Online weighting asks:

Can the chain learn which blocks are important as it runs?

We build block features w_b:

- absolute returns,
- squared returns,
- rolling volatility,
- market-wide stress,
- cross-sectional dispersion.

Then fit a streaming model:

$$\widehat{a}_{b,t}^O = \exp\{\widehat{\beta}_t^\top w_b/2\} \approx \|g_b(\theta_t)\|\,.$$

Online weighting rule

Observe sampled block-gradient feedback:

$$z_s = \log\{\|\widetilde{g}_{I_s}(\theta_s)\|^2 + \epsilon\}.$$

Discounted ridge update:

$$\widehat{\beta}_t = \arg\min_{\beta} \sum_{s<t} \rho^{t-s} \left(z_s - \beta^\top w_{I_s}\right)^2 + \tau \|\beta\|^2 .$$

Sampling rule:

$$p^O_{b,t} = (1-\lambda)\frac{\exp\{\widehat{\beta}_t^\top w_b/2\}}{\sum_r \exp\{\widehat{\beta}_t^\top w_r/2\}} + \frac{\lambda}{B}.$$

When online weighting should help

Let $a_b = \|g_b(\theta)\|$.
Uniform second moment:

$$B \sum_b a_b^2.$$

Oracle importance sampling:

$$p_b^\star \propto a_b, \qquad \text{second moment} = \left(\sum_b a_b \right)^2.$$

Uniform inefficiency:

$$\frac{B \sum_b a_b^2}{(\sum_b a_b)^2} = 1 + \mathrm{CV}(a)^2.$$

The whole game

Online weighting helps when block gradient norms are heterogeneous and predictable.

Online vs static TASS: formal comparison

Let current block importance be

$$a_{b,t} = \|g_b(\theta_t)\| \,.$$

For any sampling rule p, define the stochastic-gradient second moment

$$Q_t(p) = \sum_{b=1}^{B} \frac{a_{b,t}^2}{p_b}.$$

Exact win condition

Online is mathematically better than static TASS at time t iff

$$Q_t(p_t^O) < Q_t(p^S).$$

Interpretation

Static loses when $\widehat{a}_b^S$ is a stale proxy for today's $a_{b,t}$; online wins when $\widehat{a}_{b,t}^O$ tracks today's high-gradient blocks better.

Prediction-error view of static vs online

Ignore the small exploration floor for intuition and write

$$\widehat{a}_{b,t} = a_{b,t}\exp(\delta_{b,t}), \qquad \pi_{b,t} = \frac{a_{b,t}}{\sum_r a_{r,t}}.$$

Then the inefficiency relative to oracle is exactly

$$\frac{Q_t(\widehat{p}_t)}{Q_t(p_t^\star)} = \mathbb{E}_{\pi_t}[e^{\delta_t}]\mathbb{E}_{\pi_t}[e^{-\delta_t}].$$

Therefore online beats static when

$$\mathbb{E}_{\pi_t}[e^{\delta_t^O}]\mathbb{E}_{\pi_t}[e^{-\delta_t^O}] < \mathbb{E}_{\pi_t}[e^{\delta_t^S}]\mathbb{E}_{\pi_t}[e^{-\delta_t^S}].$$

Bounded-error sufficient condition

If $|\delta_{b,t}^O| \le \varepsilon_t^O$ and $|\delta_{b,t}^S| \le \varepsilon_t^S$, smaller current error means smaller worst-case inefficiency: $e^{2\varepsilon_t^O} < e^{2\varepsilon_t^S}$.

When should the update beat static?

Suppose current log-importance follows a drifting feature model:

$$\log a_{b,t} = \frac{1}{2}\beta_t^\top w_b + \xi_{b,t}.$$

Static TASS uses a frozen pilot $\widehat{\beta}_0$:

$$\delta_{b,t}^S \approx \frac{1}{2}(\widehat{\beta}_0 - \beta_t)^\top w_b.$$

Online uses the discounted update $\widehat{\beta}_t$:

$$\delta_{b,t}^O \approx \frac{1}{2}(\widehat{\beta}_t - \beta_t)^\top w_b.$$

Clean sufficient story

Online beats static when current estimation error is smaller than stale-model drift:

$$\sup_b |(\widehat{\beta}_t - \beta_t)^\top w_b| < \sup_b |(\widehat{\beta}_0 - \beta_t)^\top w_b|.$$

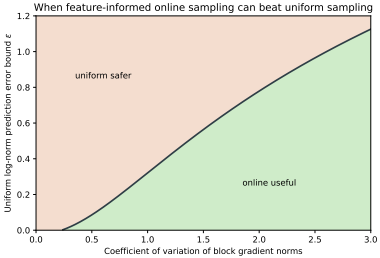
A sufficient condition

Suppose predicted gradient norms satisfy

$$e^{-\varepsilon} a_b \leq \widehat{a}_b \leq e^{\varepsilon} a_b.$$

With exploration floor λ, online beats uniform if

$$\frac{e^{2\varepsilon}}{1-\lambda} < 1 + \mathrm{CV}(a)^2.$$



Scale down to an actual crypto task

Forward-looking target

At the end of each 10-day block, predict whether the next 10-day equal-weight crypto basket return is a stress event:

$$y_b = \mathbf{1}\{R_{b+1}^{\mathrm{mkt}} \leq -7\%\}.$$

Features

Current-block return, volatility, drawdown, downside pressure, cross-asset dispersion, tail counts, and asset-specific vol/downside summaries.

This is a real crypto question

Should we be risk-on or risk-off for the next two weeks?

Why this task is covered by the math

Fit a logistic stress model:

$$h_\beta(w_b) = \sigma(\beta^\top w_b), \qquad \ell_b(\beta) = \text{cross-entropy}(y_b, h_\beta(w_b)).$$

The block gradient is

$$g_b(\beta) = \nabla \ell_b(\beta) = \{h_\beta(w_b) - y_b\} w_b.$$

Same useful-regime condition

If a few blocks are high-loss/high-gradient, and cheap crypto features predict them, online weighting should improve stochastic training.

Why static can get stale

The stress signature changes: volatility, downside skew, dispersion, liquidation-like tails. Online can track; frozen weights cannot.

Actual crypto task results

Method	Log loss	Brier	AUC	Top-risk stress rate	Strategy return	Max drawdown
Online	0.524	0.172	0.625	37.5%	-33.3%	-49.5%
Static TASS	0.531	0.175	0.623	25.0%	-25.0%	-48.3%
Uniform	0.533	0.176	0.620	25.0%	-41.5%	-55.1%
Buy-and-hold	–	–	–	–	-49.7%	-65.1%

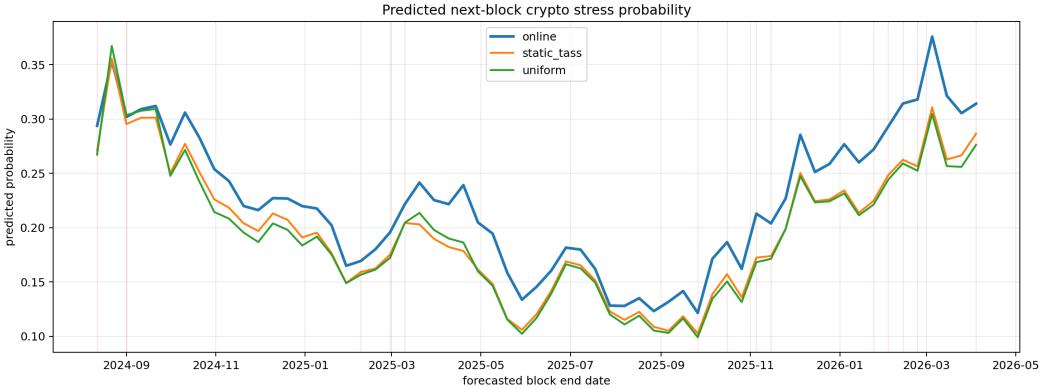
Readout

Online has the best log loss, Brier score, AUC, and top-risk stress concentration.

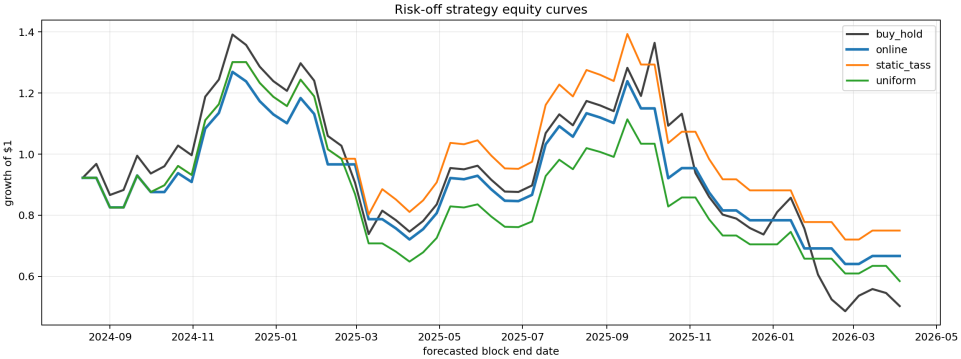
Trading caveat

Static TASS wins this simple total-return threshold rule. Online is the cleanest risk forecaster; trading P&L is a noisy downstream functional.

Online risk scores over time



Money-flavored diagnostic



Interpretation

The risk-off rule is intentionally simple: hold crypto unless predicted stress is top-quartile, then hold cash.

Then we tried to scale to full HMM inference

The hope

Use the same targeted weighting idea inside SGLD for Bayesian HMM posterior sampling.

The reality

Once the target is a full HMM posterior, gradient variance is no longer the only bottleneck.

- Label switching and multimodality.
- Weakly identified third regime.
- Transition/covariance coupling.
- Finite-step SGLD bias and numerical stability constraints.

What happened empirically

Original $K = 3$ Student-t crypto model

Collapsed into state 0 over 99% of the time. Expensive and not interpretable.

$K = 2$ rolling-standardized Gaussian model

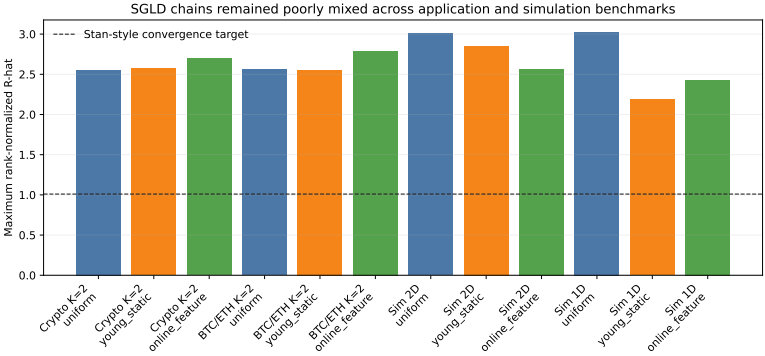
Much healthier occupancy: roughly 60/40 calm vs stress states.

But

SGLD chains still mixed poorly. Stan/NUTS mixed well for $K = 2$, and failed for $K = 3$.

The useful visual result

The painful diagnostic result



Interpretation

Good-looking state probabilities are not enough. Cross-chain diagnostics say the SGMCMC posterior samples are not trustworthy yet.

Why this does not kill the online idea

- Online weighting is a gradient-estimator variance reduction tool.
- Our failures were dominated by HMM posterior geometry:
  - label switching,
  - weakly identified third state,
  - transition/covariance coupling,
  - finite-step SGLD bias.
- If posterior geometry is the bottleneck, better block weights cannot magically fix the chain.

Cleaner claim

Online weighting deserves a benchmark where gradient-estimator variance is the bottleneck and feature-gradient relationships evolve over time.

Takeaways

1. TASS is principled: sample subsequences in proportion to parameter-specific gradient importance.
2. Online weighting helps when importance is heterogeneous, feature-predictable, and nonstationary.
3. The actual crypto stress task matches that regime; online is the best risk forecaster.
4. $K = 2$ rolling-standardized Gaussian HMM is the defensible Bayesian regime model.
5. The SGMCMC chains did not mix well enough for posterior claims.

The mooolah is not in forcing SGLD to work everywhere.
It is in using adaptivity where the math says it should pay.

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